

Modeling and Control of a Dual-Tilt Birotor UAV

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Abstract—This paper presents a modeling and control framework for a dual-tilt birotor unmanned aerial vehicle (UAV). This UAV is equipped with rotors that can tilt about two independent axes. The dual-axis tilt mechanism significantly improves vehicle maneuverability and control authority compared to conventional birotor configurations. Taking advantage of the additional degrees of actuation offered by these tilting rotors, the UAV can independently control both its position and orientation. However, the coupling between rotor speeds and tilt angles introduces nonlinearities in the mapping between control inputs and the resulting forces and moments. To address this, a constant control allocation matrix is employed to approximately map the desired force and torque commands to actuator inputs. To compensate for the mismatch between the actual and assumed control allocation, a robust backstepping control architecture is developed, ensuring system stability and tracking performance in the presence of bounded unmodeled dynamics. Numerical simulations are performed to validate the proposed control strategy in various flight scenarios, including setpoint and trajectory tracking.

I. INTRODUCTION

Quadrotors are the most commonly used multirotor unmanned aerial vehicles (UAVs) and have gained widespread popularity in various aerial applications due to their mechanical simplicity and agility [1]. However, they exhibit certain limitations, particularly high power consumption and a relatively large structural footprint, which constrain their effectiveness in confined or cluttered environments [2].

To address these issues, researchers have recently turned their attention to birotor UAVs [3]. Owing to their reduced geometric dimensions and simpler mechanical design, birotors offer a more compact alternative, making them well-suited for missions that require navigation through narrow or constrained spaces, such as indoor inspection, search-and-rescue in collapsed structures and payload capabilities [4], [5], [6]. Existing literature has mainly focused on designing robust controller for such birotor configuration UAVs [7], [8], [9], [10], [11]. Typically, such birotor UAVs feature rotor tilting about a single axis, resulting in only four actuator inputs to control a six-degree-of-freedom (6-DoF) system [3], [5], [6].

Despite their compactness and lower power requirements, conventional single-axis tilt birotors remain underactuated and lack the capability to independently control both position and orientation [6], [7], [8], [9], [10]. This limits maneuverability in confined environments, such as pipeline

inspections, where the vehicle must maintain a specific orientation while moving [12].

To address the problem of under-actuation for existing birotor configuration, we propose a dual tilt-enabled mechanism that allows each rotor to tilt about two independent axes. This configuration significantly improves vehicle maneuverability by allowing independent control over all six degrees of freedom (DoF), effectively transforming the birotor into a fully actuated system.

This paper presents a comprehensive modeling and control framework for a fully actuated dual-tilt birotor UAV (DTB-UAV). A key challenge addressed in this work is the nonlinear coupling between actuator inputs to the control forces and moments. To address this, a constant control allocation matrix is used to approximately map the desired force and torque commands to actuator inputs. The use of a constant allocation matrix eases real-time implementation and reduces computational load. To mitigate the error between the actual and assumed constant allocation matrices, a robust backstepping control architecture is developed, ensuring system stability and tracking performance in the presence of bounded unmodeled dynamics. The key contributions of this work are as follows:

- We derive the mathematical model of the dual-axis tilt augmented birotor UAV. This is different from the existing birotors as their rotors tilt about single axis [7].
- We demonstrated that the DTB-UAV can achieve independent control of both position and orientation, as compared to conventional birotor configurations.
- We design of a robust backstepping control architecture that compensates for the mismatch between the actual and assumed constant control allocation, ensuring bounded tracking error in the presence of bounded unmodeled dynamics. The use of a constant allocation matrix improves computational efficiency by avoiding the need to compute the inverse of a state-dependent matrix.

The organization of the paper is as follows: Section II outlines the mathematical model for the DBT-UAV. Section III discusses how the dual-tilt birotor can independently control its position and orientation. Section IV discusses the robust control design and efficient control allocation scheme for the dual-tilt birotor. The simulation results are presented in Section V, and a summary of the work can be found in Section VI.

A. Abbreviations and Acronyms

Throughout this paper, $\mathcal{F}_b$ and $\mathcal{F}_I$ denote the body frame and the inertial frame, respectively. The orientation of the

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vehicle frame $\mathcal{F}_b$ with respect to the inertial frame $\mathcal{F}_I$ is represented by a rotation matrix $\mathcal{R}$ in terms of Euler angles (ϕ, θ, ψ) is given in [13], where ϕ, θ, ψ represent the roll, pitch and yaw respectively.

II. MATHEMATICAL MODELING

This section presents the mathematical model of a DTB-UAV.

A. Position and Attitude Dynamics

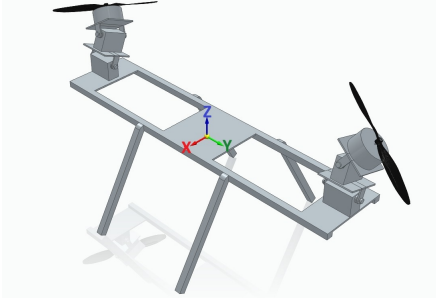


Fig. 1: CAD model of a dual-tilt birotor

The vehicle has been modeled as a rigid body. Figure 1 shows the CAD model of the DTB-UAV. The dynamics of the dual-tilt birotor are derived using the Newton–Euler equations of motion [2], [12], and expressed as:

$$\ddot{\mathbf{p}} = \mathcal{R} \frac{\mathbf{F}_b}{m} - \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix}, \quad (1a)$$

$$\mathcal{J} \dot{\boldsymbol{\omega}}_b = -\boldsymbol{\omega}_b \times (\mathcal{J} \boldsymbol{\omega}_b) + \mathbf{M}_b, \quad (1b)$$

where $\times$ denotes the cross product between vectors in $\mathbb{R}^3$, $\mathbf{F}_b = [F_{b1}, F_{b2}, F_{b3}]^\top$ and $\mathbf{M}_b = [M_{b1}, M_{b2}, M_{b3}]^\top$ are the control force and moment vectors in $\mathcal{F}_b$, $\boldsymbol{\omega}_b$ is the angular velocity vector of the body in $\mathcal{F}_b$, $\mathbf{p}$ is the position of the vehicle's center of gravity (CoG) with respect to the inertial frame $\mathcal{F}_I$, m is the mass of the vehicle, and $\mathcal{J}$ is the moment of inertia matrix of the vehicle $\mathcal{F}_b$. The relation between Euler angle rates and angular velocity [13] is defined as follows:

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & \sin \phi \tan \theta & \cos \phi \tan \theta \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi \sec \theta & \cos \phi \sec \theta \end{bmatrix} \boldsymbol{\omega}_b = \mathcal{T}(\phi, \theta) \boldsymbol{\omega}_b \quad (2)$$

where ϕ, θ, ψ represent the roll, pitch, and yaw respectively.

B. Control Force and Torque

Let Ω_i denote the angular speed of the i^{th} rotor. The thrust force and drag moment generated by the i^{th} rotor are given by $F_i = k_m \Omega_i^2$ and $M_i = k_d \Omega_i^2$, respectively, where k_m and k_d denote the thrust and drag coefficients. In a dual-tilt birotor configuration, each rotor is first tilted about the X -axis (by angle β_i), followed by a tilt about the Y -axis (by angle α_i). The net thrust vector generated by the i^{th} rotor ($\mathbf{T}_i = [T_{x,i}, T_{y,i}, T_{z,i}]^\top$) in the body frame $\mathcal{F}_b$ is given by:

$$\mathbf{T}_i = \begin{bmatrix} F_i \cos \beta_i \sin \alpha_i \\ -F_i \sin \beta_i \\ F_i \cos \beta_i \cos \alpha_i \end{bmatrix} \quad (3)$$

The total control force in $\mathcal{F}_b$ is:

$$\mathbf{F}_b = \mathbf{T}_1 + \mathbf{T}_2. \quad (4)$$

Similarly, the drag moment vector generated by the i^{th} rotor in $\mathcal{F}_b$ is:

$$\mathbf{N}_i = \begin{bmatrix} M_i \cos \beta_i \sin \alpha_i \\ -M_i \sin \beta_i \\ M_i \cos \beta_i \cos \alpha_i \end{bmatrix} = \frac{k_d}{k_m} \mathbf{T}_i. \quad (5)$$

From Fig.2, the position vector from the CoG to the rotor

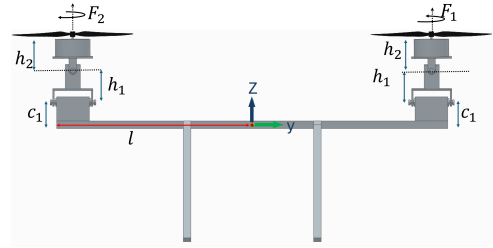


Fig. 2: Position of the rotors with respect to CoG

1, denoted by $\mathbf{r}_1$ in $\mathcal{F}_b$, is given by:

$$\mathbf{r}_1 = \begin{bmatrix} (h_1 + h_2 \cos \beta_1) \sin \alpha_1 \\ l - h_2 \sin \beta_1 \\ c_1 + (h_1 + h_2 \cos \beta_1) \cos \alpha_1 \end{bmatrix} = \begin{bmatrix} r_{1x} \\ r_{1y} \\ r_{1z} \end{bmatrix}, \quad (6)$$

where c_1 denotes the vertical distance from the center of gravity (CoG) to the first tilt axis, h_1 represents the distance between the first and second tilt axes, and h_2 corresponds to the distance from the propeller to the second tilt axis, as illustrated in Fig. 2. Similarly, for rotor 2, the distance from the CoG, denoted by $\mathbf{r}_2$ in $\mathcal{F}_b$, is given by:

$$\mathbf{r}_2 = \begin{bmatrix} (h_1 + h_2 \cos \beta_2) \sin \alpha_2 \\ -(l + h_2 \sin \beta_2) \\ c_1 + (h_1 + h_2 \cos \beta_2) \cos \alpha_2 \end{bmatrix} = \begin{bmatrix} r_{2x} \\ r_{2y} \\ r_{2z} \end{bmatrix}. \quad (7)$$

Since rotor 1 rotates clockwise and rotor 2 rotates counterclockwise, the net control moment $\mathbf{M}_b$ in $\mathcal{F}_b$ is given by:

$$\mathbf{M}_b = -\frac{k_d}{k_m} \mathbf{T}_1 + \frac{k_d}{k_m} \mathbf{T}_2 + \mathbf{r}_1 \times \mathbf{T}_1 + \mathbf{r}_2 \times \mathbf{T}_2. \quad (8)$$

The next section discusses how the DTB-UAV can independently control its position and orientation based on the system dynamics.

III. INDEPENDENT CONTROL FOR DBT-UAV

Conventional birotors are underactuated systems, as they have four actuators but six degrees of freedom. To demonstrate that the DTB-UAV can independently control both its attitude and position, it must be capable of generating force and torque in all directions [14]. Let $\mathcal{C}$ denote the mapping from the thrust generated by each rotor to the resulting force and torque acting on the vehicle. For the

system to generate arbitrary force and torque vectors, the matrix $\mathcal{C}$ must be full rank. Let $\mathbf{f}$ be a 6×1 vector that collects the thrust components generated by both rotors along the body-frame axes. These total thrust vector $\mathbf{f}$ is given as $\mathbf{f} = [T_{x,1}, T_{y,1}, T_{z,1}, T_{x,2}, T_{y,2}, T_{z,2}]^\top$. From (4) and (8), $\mathbf{F}_b$ and $\mathbf{M}_b$ can be expressed in a relation to $\mathbf{f}$ as

$$\begin{bmatrix} \mathbf{F}_b \\ \mathbf{M}_b \end{bmatrix} = \mathcal{C}(\alpha_i, \beta_i) \mathbf{f}. \quad (9)$$

where $\mathcal{C}(\alpha_i, \beta_i)$ is given by

$$\mathcal{C}(\alpha_i, \beta_i) = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ -\frac{k_d}{k_m} & -r_{1z} & r_{1y} & \frac{k_d}{k_m} & -r_{2z} & r_{2y} \\ r_{1z} & \frac{k_d}{k_m} & -r_{1x} & r_{2z} & -\frac{k_d}{k_m} & -r_{2x} \\ -r_{1y} & r_{1x} & -\frac{k_d}{k_m} & -r_{2y} & r_{2x} & \frac{k_d}{k_m} \end{bmatrix} \quad (10)$$

From (6), (7), and (10), each column of the matrix $\mathcal{C}$ is linearly independent. Therefore, $\mathcal{C}$ is full rank for all values of the tilt angles, i.e., $\forall \alpha_1, \forall \beta_1, \forall \alpha_2, \forall \beta_2$. As a result, the proposed vehicle can independently generate force and torque in all directions. This property allows us to decompose the system into two subsystems position and attitude—and design the control laws for each independently, as illustrated in Fig. 3. The next section presents the control design methodology with control allocation for the DTB-UAV.

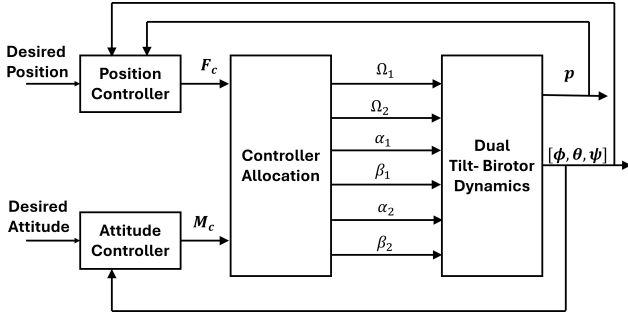


Fig. 3: Overall Controller Block Diagram

IV. CONTROL DESIGN FOR A DTB-UAV

In this work, we aim to find a simple, constant control allocation matrix that can be hard-coded into the controller for real-time implementation. To achieve this, we consider a nominal control allocation matrix, where the tilt angles are fixed as $\alpha_i = 0$ and $\beta_i = 0$. Under this assumption, a constant control allocation matrix $\mathcal{C}_0 = \mathcal{C}(0, 0)$, is computed.

However, in practice, the tilt angles (α_i, β_i) will vary during flight, resulting in a deviation between the actual control allocation $\mathcal{C}(\alpha_i, \beta_i)$ and nominal control allocation $\mathcal{C}_0$. The difference between the actual and nominal control allocation matrices is defined as

$$\Delta \mathcal{C} = \mathcal{C}(\alpha_i, \beta_i) - \mathcal{C}_0. \quad (11)$$

Using (11) in (9), we obtain:

$$\begin{bmatrix} \mathbf{F}_b \\ \mathbf{M}_b \end{bmatrix} = \mathcal{C}_0 \mathbf{f} + \Delta \mathcal{C} \mathbf{f} = \begin{bmatrix} \mathbf{F}_c \\ \mathbf{M}_c \end{bmatrix} + \begin{bmatrix} \mathbf{F}_d \\ \mathbf{M}_d \end{bmatrix}, \quad (12)$$

where $\mathbf{F}_c$ and $\mathbf{M}_c$ represent the nominal control force and moment computed using the constant allocation matrix $\mathcal{C}_0$, and $\mathbf{F}_d$ and $\mathbf{M}_d$ represent the disturbance force and moment due to unmodeled dynamics.

The term $\Delta \mathcal{C} \mathbf{f} = [\mathbf{F}_d^\top \mathbf{M}_d^\top]^\top$ captures the deviation between the nominal and actual control allocation maps. By evaluating $\Delta \mathcal{C} \mathbf{f}$ over the entire operational range of tilt angles (α_i, β_i) and the maximum allowable rotor thrust, we estimate the maximum possible deviation. This bounded unmodeled dynamics must be explicitly accounted for during controller design to ensure robustness and stability. Therefore, the design of a robust controller is essential to account for this modeling uncertainty and to guarantee stable and reliable flight performance of the DTB-UAV under all feasible operating conditions.

A. Robust Controller for Attitude Subsystem

The objective of this section is to design the control input $\mathbf{M}_c$ such that the Euler angles $\mathbf{r} := [\phi, \theta, \psi]^\top$ converge to the desired trajectory $\mathbf{r}_d := [\phi_d, \theta_d, \psi_d]^\top$. To achieve this, a robust backstepping control strategy is developed. Let the attitude tracking error be defined as

$$\mathbf{r}_e = \mathbf{r} - \mathbf{r}_d. \quad (13)$$

A positive definite Lyapunov function in terms of $\mathbf{r}_e$ is given by

$$V(\mathbf{r}_e) = \frac{1}{2} \mathbf{r}_e^\top \mathbf{r}_e. \quad (14)$$

The time derivative of (14), using (2), is

$$\dot{V}(\mathbf{r}_e) = \mathbf{r}_e^\top (\dot{\mathbf{r}}_d - \mathcal{T} \boldsymbol{\omega}_b). \quad (15)$$

Assuming $\boldsymbol{\omega}_b$ as a virtual control input, a control law is proposed as

$$\boldsymbol{\omega}_b = \mathcal{T}^{-1}(\dot{\mathbf{r}}_d + K_1 \mathbf{r}_e), \quad (16)$$

where $K_1 \in \mathbb{R}^{3 \times 3}$ is a positive definite diagonal gain matrix. If the condition in (16) is satisfied, then $\dot{V}(\mathbf{r}_e)$ is negative definite, implying that $\mathbf{r}_e$ converges asymptotically to the origin. However, $\boldsymbol{\omega}_b$ is a system state rather than a control input. Therefore, its desired value is defined as

$$\boldsymbol{\omega}_d = \mathcal{T}^{-1}(\dot{\mathbf{r}}_d + K_1 \mathbf{r}_e). \quad (17)$$

We then design a control law for $\mathbf{M}_c$ such that the actual angular velocity $\boldsymbol{\omega}_b$ tracks $\boldsymbol{\omega}_d$. Define the angular velocity tracking error as $\boldsymbol{\omega}_e := \boldsymbol{\omega}_d - \boldsymbol{\omega}_b$. To ensure that the tracking errors $\mathbf{r}_e$ and $\boldsymbol{\omega}_e$ remain uniformly bounded in the presence of model uncertainty, a robust attitude control law is proposed. From Section IV, $\mathbf{M}_d$ is bounded by $\|\mathbf{M}_d\|_2 \leq b_a$. The robust control law is given by:

$$\mathbf{M}_c = K_2 \boldsymbol{\omega}_e + \mathcal{T} \mathbf{r}_e + \boldsymbol{\omega}_b \times \mathcal{J} \boldsymbol{\omega}_b + \mathcal{J} \boldsymbol{\omega}_d + \boldsymbol{\nu}_r, \quad (18)$$

where $K_2 \in \mathbb{R}^{3 \times 3}$ is a positive definite diagonal gain matrix, and $\boldsymbol{\nu}_r$ is defined as

$$\boldsymbol{\nu}_r = \frac{b_a^2 \boldsymbol{\omega}_e}{b_a \|\boldsymbol{\omega}_e\|_2 + \gamma_r}, \quad (19)$$

with $\gamma_r > 0$ being a small positive constant to ensure smoothness and avoid division by zero.

Theorem 1: Consider the subsystem described by (1b) and (12), with $\|\mathbf{M}_d\|_2 \leq b_a$. Under the control law (18), the attitude error $\mathbf{r}_e$ and angular velocity error $\boldsymbol{\omega}_e$ remain bounded where the bounds are given by:

$$\|\mathbf{r}_e\|_2 \leq \sqrt{\frac{\gamma_r}{d_r}}, \|\boldsymbol{\omega}_e\|_2 \leq \sqrt{\frac{\gamma_r}{d_r}}, \quad (20)$$

with $d_r = \min\{\lambda_{\min}(K_1), \lambda_{\min}(K_2)\}$.

Proof: Consider the following Lyapunov candidate:

$$V_{a1} = \frac{1}{2} \boldsymbol{\omega}_e^\top \mathcal{J} \boldsymbol{\omega}_e + \frac{1}{2} \mathbf{r}_e^\top \mathbf{r}_e. \quad (21)$$

Taking the time derivative of V_{a1} and substituting from (1b), we get:

$$\begin{aligned} \dot{V}_{a1} = & \boldsymbol{\omega}_e^\top \mathcal{J} \dot{\boldsymbol{\omega}}_d - \boldsymbol{\omega}_e^\top (\mathbf{M}_c + \mathbf{M}_d) + \boldsymbol{\omega}_e^\top (\boldsymbol{\omega} \times \mathcal{J} \boldsymbol{\omega}) \\ & + \mathbf{r}_e^\top \mathcal{T} \boldsymbol{\omega}_e - \mathbf{r}_e^\top K_1 \mathbf{r}_e. \end{aligned} \quad (22)$$

Substituting the control law from (18), we obtain:

$$\dot{V}_{a1} = -\boldsymbol{\omega}_e^\top K_2 \boldsymbol{\omega}_e + \boldsymbol{\omega}_e^\top (-\boldsymbol{\nu}_r + \mathbf{M}_d) - \mathbf{r}_e^\top K_1 \mathbf{r}_e. \quad (23)$$

Now, the cross-term $\boldsymbol{\omega}_e^\top (-\boldsymbol{\nu}_r + \mathbf{M}_d)$ satisfies:

$$\boldsymbol{\omega}_e^\top (-\boldsymbol{\nu}_r + \mathbf{M}_d) \leq \|\boldsymbol{\omega}_e\|_2 \|\mathbf{M}_d\|_2 - \boldsymbol{\omega}_e^\top \boldsymbol{\nu}_r.$$

Given the bound $\|\mathbf{M}_d\|_2 \leq b_a$ and the definition of $\boldsymbol{\nu}_r$,

$$\boldsymbol{\omega}_e^\top (-\boldsymbol{\nu}_r + \mathbf{M}_d) \leq \gamma_r \frac{b_a \|\boldsymbol{\omega}_e\|_2}{b_a \|\boldsymbol{\omega}_e\|_2 + \gamma_r} \leq \gamma_r. \quad (24)$$

Substituting (24) into (23), we have:

$$\begin{aligned} \dot{V}_{a1} \leq & -\boldsymbol{\omega}_e^\top K_2 \boldsymbol{\omega}_e - \mathbf{r}_e^\top K_1 \mathbf{r}_e + \gamma_r \\ \leq & -d_r (\|\boldsymbol{\omega}_e\|_2^2 + \|\mathbf{r}_e\|_2^2) + \gamma_r, \end{aligned} \quad (25)$$

where $d_r = \min\{\lambda_{\min}(K_1), \lambda_{\min}(K_2)\}$.

The inequality in (25) implies that V_{a1} is negative definite outside the compact set:

$$S_r = \left\{ (\mathbf{r}_e, \boldsymbol{\omega}_e) \mid \|\boldsymbol{\omega}_e\|_2^2 + \|\mathbf{r}_e\|_2^2 \leq \frac{\gamma_r}{d_r} \right\}. \quad (26)$$

Since V_{a1} is a radially unbounded, positive definite function, the error trajectories $\mathbf{r}_e(t)$ and $\boldsymbol{\omega}_e(t)$ decrease whenever they are outside the set S_r . This proves that $\mathbf{r}_e$ and $\boldsymbol{\omega}_e$ are bounded, with bounds as stated in (20). ■

Remark 1: The tracking error bound given in (20) can be reduced by increasing the controller gains K_1 and K_2 or reducing γ_r . However, from the control law in (18), this reduction in the error bound results in an increase in control effort. Therefore, a trade-off must be considered between improving tracking accuracy and avoiding actuator saturation during controller tuning.

B. Robust Controller for Position Subsystem

The objective is to design the control law $\mathbf{F}_c$ such that the vehicle tracks the desired position trajectory. Let the desired position of the CoG of the DTB in the inertial frame be denoted by $\mathbf{p}_d = [x_d, y_d, z_d]^\top$. Define the position tracking error as $\mathbf{p}_e = \mathbf{p}_d - \mathbf{p}$. Similar to the controller designed for the attitude subsystem discussed in Section IV-A, the

robust control law for the position subsystem is given by the following:

$$\mathbf{F}_c = \mathcal{R}^\top (K_4 \mathbf{v}_e + m[0, 0, g]^\top + \mathbf{p}_e + m\dot{\mathbf{v}}_d + \boldsymbol{\nu}_p), \quad (27)$$

where $\mathbf{v}_e = \mathbf{v}_d - \dot{\mathbf{p}}$ is the velocity tracking error in $\mathcal{F}_I$, and the desired velocity $\mathbf{v}_d$ is defined as $\mathbf{v}_d = K_3 \mathbf{p}_e + \dot{\mathbf{p}}_d$, with $K_3 \in \mathbb{R}^{3 \times 3}$ and $K_4 \in \mathbb{R}^{3 \times 3}$ being positive definite gain matrices. The robust term $\boldsymbol{\nu}_p$ is defined as

$$\boldsymbol{\nu}_p = \frac{b_p^2 \mathbf{v}_e}{b_p \|\mathbf{v}_e\|_2 + \gamma_p}, \quad (28)$$

where $\gamma_p > 0$ is a small design constant, and b_p denotes the upper bound of the disturbance such that $\|\mathbf{F}_d\|_2 \leq b_p$.

Theorem 2: Consider the subsystem described by (1a) and (12), with $\|\mathbf{F}_d\|_2 \leq b_p$. Under the control law (27), the position error $\mathbf{p}_e$ and velocity error $\mathbf{v}_e$ remain bounded where the bounds are given by

$$\|\mathbf{p}_e\|_2 \leq \sqrt{\frac{\gamma_p}{d_p}}, \|\mathbf{v}_e\|_2 \leq \sqrt{\frac{\gamma_p}{d_p}}, \quad (29)$$

with $d_p = \min\{\lambda_{\min}(K_3), \lambda_{\min}(K_4)\}$.

Proof: Consider the Lyapunov candidate:

$$V_p = \frac{1}{2} \mathbf{v}_e^\top m \mathbf{v}_e + \frac{1}{2} \mathbf{p}_e^\top \mathbf{p}_e. \quad (30)$$

Following the same procedure as in the proof of Theorem 1, the error trajectories $(\mathbf{p}_e, \mathbf{v}_e)$ are bounded, with bounds given in (29). ■

C. Control Allocation

Given the control force $\mathbf{F}_c$ and control moment $\mathbf{M}_c$, the objective is to compute the actuator inputs, the rotor speeds (Ω_i) and tilt angles (α_i, β_i). From (12), the total thrust vector $\mathbf{f}$ is computed using the nominal control allocation matrix $\mathcal{C}_0$ as $\mathbf{f} = \mathcal{C}_0^{-1} [\mathbf{F}_c, \mathbf{M}_c]^\top$. Since $\mathcal{C}_0$ is a constant control allocation matrix, its inverse is computed offline once to reduce computational complexity. Using (3), the actuator inputs for the i^{th} rotor are then computed as:

$$\begin{aligned} \Omega_i &= \left(\frac{f_{x,i}^2 + f_{y,i}^2 + f_{z,i}^2}{k_m^2} \right)^{\frac{1}{4}}, \\ \alpha_i &= \arctan(f_{x,i}, f_{z,i}), \\ \beta_i &= \arctan(-f_{x,i}, \sqrt{f_{x,i}^2 + f_{z,i}^2}), \end{aligned} \quad (31)$$

These expressions provide a direct mapping from the desired control inputs to the physical actuator commands for the DTB-UAV.

V. RESULTS

In this section, two sets of simulations were performed to validate the proposed controller with the control allocation for the DTB-UAV. The model parameter for the dual-tilt birotor used in the simulation is given in Table I. Depending on the CAD design, the maximum tilt angle for the dual-tilt birotor was found to be $\pm 30^\circ$, chosen to ensure that the propellers do not collide with the arms of the vehicle. The maximum thrust generated by each rotor is assumed

TABLE I: Dual-tilt Birotor Parameter used for Simulation

Parameters	Values
Moment-of-inertia $\mathcal{J}$	$\text{diag}([0.015, 0.015, 0.03]) \text{ kg/m}^2$
Length of the arm l	30 cm
Height of the servo to base c_1	3 cm
Distance from one servo to another h_1	3 cm
Distance from the servo to the rotor tip h_2	5 cm
Force coefficient by drag coefficient $\frac{k_f}{k_m}$	0.017 m^{-1}
Mass of the tilt-augmented quadrotor m	1.5 kg

to be mg , resulting in a combined thrust of $2mg$ for the vehicle. Based on these constraints, the upper bounds on $\mathbf{F}_d$ and $\mathbf{M}_d$, and the controller gains used in this work are given in Table. II. In this section, Matlab simulations

TABLE II: Controller gain values

Parameters	Values	Parameters	Values
K_1	$\text{diag}([5, 5, 5])$	γ_r	0.01
K_2	$\text{diag}([4, 4, 4])$	γ_p	0.1
K_3	$\text{diag}([3, 3, 3])$	b_r	0.3
K_4	$0.2\text{diag}([1, 1, 1])$	b_p	0.2

are performed to demonstrate the tracking performance of the proposed controller. The dual-tilt birotor is simulated in MATLAB with model parameters given in Table. I and control parameters given in Table. II.

A. Setpoint Tracking

To evaluate the performance of the proposed controller, a step command is applied to both the attitude and position subsystems. The vehicle is commanded to sequentially follow a set of predefined waypoints that form the corners of a rectangular trajectory in 3D space. The four waypoints are defined as A $([0, 0, 5])$, B $([0, 1, 5])$, C $([1, 1, 5])$, and D $([1, 0, 5])$, as illustrated in Fig. 4.

Once the vehicle reaches a safe altitude from the ground, the desired attitude is set to $[\phi_d, \theta_d, \psi_d] = [10^\circ, -10^\circ, 45^\circ]$. The performance of attitude tracking using the proposed robust controller and constant control allocation scheme is shown in Fig. 5. The vehicle settles within a tolerance band $\pm 2^\circ$ for both roll and pitch, satisfying the maximum bound specified in (20) within approximately 3 seconds. Minor deviations in roll and pitch are attributed to the mismatch between the nominal (assumed) and actual control allocation matrices. The position tracking response is presented in Fig. 4. The vehicle successfully follows the desired waypoints, with a maximum deviation of approximately $\pm 0.15 \text{ m}$ from the reference positions, satisfying the bound specified in (29). The DBT-UAV takes around 10 seconds to transition between consecutive waypoints. The corresponding rotor thrust inputs and tilt angles are shown in Fig. 6. Sharp changes in thrust and tilt angles are observed during transitions between waypoints, which correspond to step changes in the position and attitude commands. It is also observed that all control inputs remain well within the

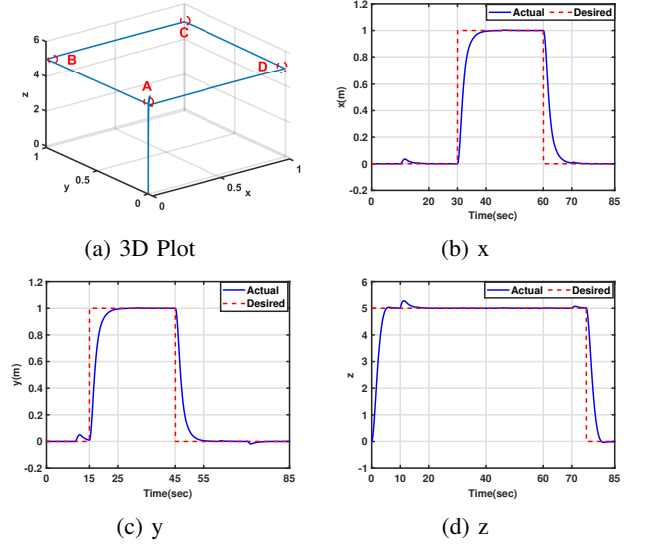


Fig. 4: Position tracking of a DTB-UAV for step command

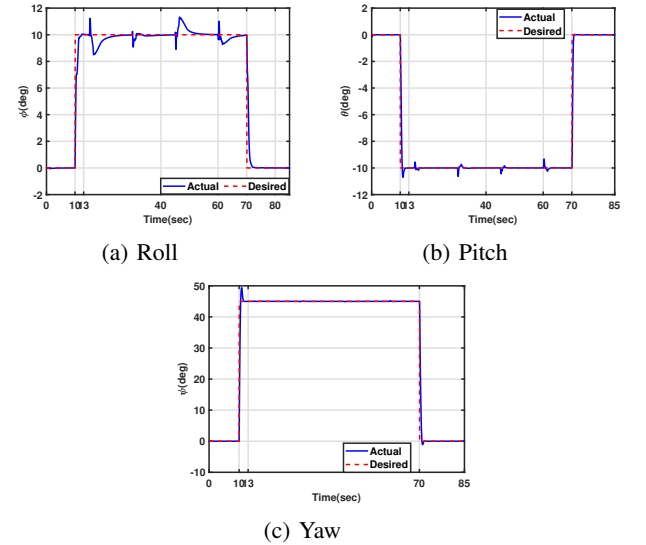


Fig. 5: Attitude tracking of a DTB-UAV for step command

actuator limits. These results validate the robustness and effectiveness of the proposed controller in tracking both attitude and position commands, even in the presence of bounded modeling uncertainties.

B. Helical Trajectory Control

In the second set of simulations, the dual-tilt birotor is commanded to follow a helical trajectory while simultaneously tracking a time-varying attitude. The desired position for the helical trajectory is given by $\mathbf{p}_d = [\sin(2\pi t/40), \cos(2\pi t/40), 0.2t]^\top$ and the desired attitude is given by $[\phi_d, \theta_d, \psi_d]^\top = [15 \sin(2\pi t/20), -15 \sin(2\pi t/20), 45]^\top$ degrees for $t \geq 20$ seconds. The tracking performance of both the position and attitude subsystems is illustrated in Fig. 7 and Fig. 8, respectively. The vehicle is observed to successfully follow

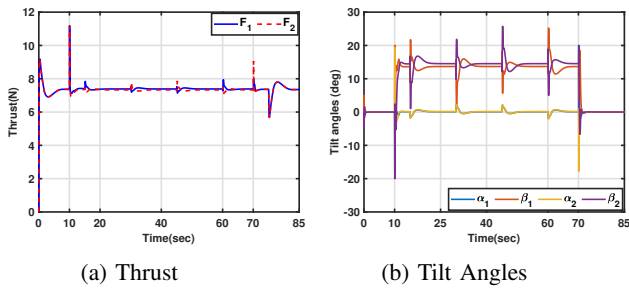


Fig. 6: Rotor thrust and Tilt angles for step command

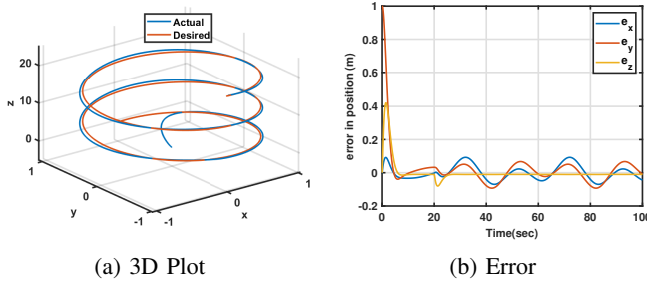


Fig. 7: Position trajectory and corresponding position error of a DTB-UAV for helical trajectory command

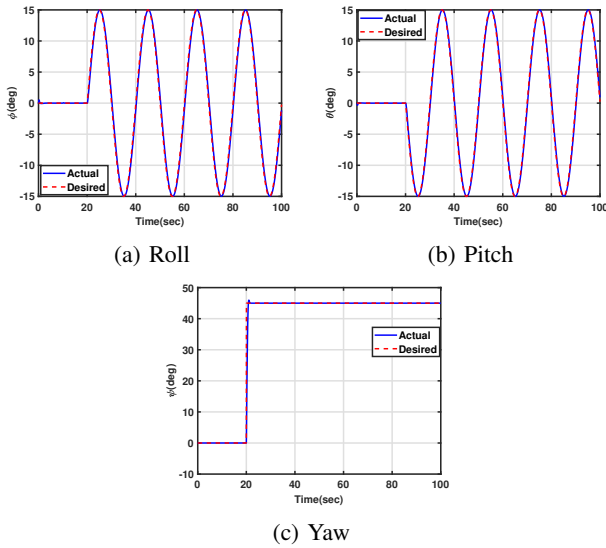


Fig. 8: Attitude tracking of a DTB-UAV for helical trajectory command

the desired trajectory while maintaining decoupled control over position and attitude. Position and attitude tracking errors remain within 0.15 m and 2° , respectively, throughout the maneuver. These results validate the effectiveness of the proposed robust control strategy in conjunction with a constant control allocation matrix, demonstrating independent tracking of both translational and rotational commands.

VI. CONCLUSIONS

This paper presented the modeling and control design of a novel dual-tilt birotor UAV (DTB-UAV), which utilizes rotors capable of tilting about two independent axes. The dual axis tilt mechanism enhances the maneuverability of the platform and allows full control over both position and orientation, overcoming the inherent limitations of conventional birotor designs. A computationally efficient control allocation strategy was proposed to map desired force and torque commands to actuator input. To account for the effects of allocation errors, a robust backstepping control framework was designed, ensuring stability and convergence to a desired bounded region in the presence of bounded unmodeled dynamics. The effectiveness of the proposed approach was validated through numerical simulations, demonstrating set-point tracking, and trajectory following. This work lays the foundation for further experimental validation.

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